Kashish Lalwani

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Data & AI Engineer building evaluation-driven ML systems and KPI-governed decision frameworks that translate large-scale data into reliable, measurable impact. Experienced across retrieval systems, experimentation infrastructure, product analytics, and cost optimization, bridging technical execution with data-driven strategic decision-making in production environments.

SKILLS

Languages, Databases & Cloud: Python, SQL, R, C/C++, PostgreSQL, MySQL, BigQuery, AWS
ML, AI & GenAI: Pandas, NumPy, scikit-learn, TensorFlow, PyTorch, Transformers, RAG, Embeddings, FAISS, LangChain, GPT APIs
Data & Analytics: A/B Testing, KPI Design, Statistical Testing, Feature Engineering, Model Evaluation, Root-Cause Analysis
Frameworks & Tools: FastAPI, Airflow, Docker, Git, Tableau, Power BI, Looker, Excel

EXPERIENCE

Research Assistant, Stony Brook University, USA **Aug 2025 – Dec 2025**
Python | SQL | Retrieval Systems | Statistical Evaluation

- Engineered deterministic ML evaluation framework processing 1M+ records and 100K+ queries to benchmark robustness across multi-engine retrieval systems.
- Designed reproducible cross-engine pipelines with cached artifacts and validation logic, eliminating metric inconsistencies and strengthening KPI credibility for system comparison.
- Implemented True-PIR–based robustness evaluation and statistical bias diagnostics, enabling systematic detection of semantic drift and ranking instability under query perturbations.
- Automated audit and reporting workflows improving visibility into system behavior and reducing manual evaluation overhead.
- Profiled latency, throughput, and scaling constraints to identify bottlenecks and guide retrieval optimization under large-scale inference loads.

Founding AI Engineer, Versaunt (Aayats), Inc., USA **May 2025 – Aug 2025**
ML Systems | Product Analytics | Experimentation

- Designed and productionized experimentation infrastructure processing 50K+ daily events to monitor model behavior across 10K+ users and 500K+ monthly queries.
- Built behavioral feature pipelines over 1M+ interaction records, improving signal quality, evaluation robustness, and real-world performance consistency.
- Owned KPI dashboards and evaluation pipelines influencing roadmap prioritization and release validation decisions.
- Implemented A/B experimentation workflows reducing experiment turnaround 30% and accelerating validated model releases.
- Integrated API-based inference pipelines with automated monitoring, detecting degradation patterns and strengthening production stability.

Data Analyst Intern, Saroj Group of Companies, IN **Jan 2024 – July 2024**
Operational Analytics | Predictive Modeling

- Reduced defect rate 40% using statistical modeling on 100K+ records, generating $50K annual savings through root-cause-driven process optimization.
- Improved profit margin 12% by identifying cost leakage and operational inefficiencies through structured SQL and Python analysis.
- Automated reporting pipelines eliminating 25+ manual hours per month and improving data reliability for operational planning.

PROJECTS

Scalable Retrieval Benchmarking for RAG Systems
Python | FAISS | Vector Embeddings | NumPy | Retrieval Evaluation | Latency | Throughput Profiling

- Built dense embedding and FAISS indexing pipelines simulating 10K–50K chunk workloads; profiled p50/p95 latency and throughput to evaluate scalability trade-offs.
- Compared embedding and indexing strategies to guide retrieval performance optimization under scaling constraints.

RAG Financial Intelligence Platform
Python | LangChain | GPT APIs | Vector Search | FAISS | Embeddings | Retrieval-Augmented Generation | ETL Pipelines

- Designed end-to-end RAG pipeline (chunking → embedding → vector search → generation) supporting 20+ document types for structured financial analysis.
- Optimized retrieval and inference flow, improving analysis speed 60% and decision efficiency 40% through latency-aware pipeline design.

AI-Enhanced Portfolio Optimization System | *Python | Pandas | Statistical Modeling | Risk Modeling (CVaR) | Time-Series Analysis*

- Developed CVaR-based portfolio optimization model improving Sharpe ratio 15% and reducing downside risk 28% using risk-adjusted modeling.
- Automated real-time data ingestion enabling dynamic portfolio rebalancing and faster decision response.

Formula 1 Performance Intelligence System | *SQL | Python | Pandas | KPI Analytics | Data Visualization | Tableau | Power BI*

- Built SQL-driven analytics platform analyzing 150K+ telemetry records; designed 20+ KPIs for benchmarking and performance monitoring dashboards.

EDUCATION

Stony Brook University – *Master of Science in Data Science* **Expected May 2026**
Shivaji University – *B.Tech in Computer Science (AI & ML)* **Aug 2020 – May 2024**
Honors: Best Outgoing Student — University Distinction